

A Reflexive Case of the L^p State-Space Problem

Automatic Math Discovery Smoke Test

June 9, 2026

Source problem

The source paper asks for an analogue, for $1 < p < \infty$, of its characterization of weakly compact state spaces in $L^1(\mu, X)$.

not a very smooth point (see, [11, Proposition 2.3]).

3.2. Open Problems: In what follows, we present some open problems.

Problem 3.17. *In Theorems 2.5 and 2.8, we studied the w^* - w and w^* - $\|\cdot\|$ points of continuity of the state spaces. However, a characterization of w^* - w points of continuity of the dual unit ball $\ell^q(X^*)_1$ is not known. In particular, even for a norm-attaining functional $(x_i^*) \in \ell^q(X^*)$, the conclusion remains open.*

Problem 3.18. *In Theorem 3.6, we have presented a characterization of weakly compact state spaces in the space $L^1(\mu, X)$. A similar result is not known for the space $L^p(\mu, X)$, where $1 < p < \infty$.*

Result

This packet gives the full reflexive-domain answer and shows that the L^1 singleton phenomenon cannot extend to $p > 1$.

Theorem 1. *Let (Ω, Σ, μ) be a finite measure space, let $1 < p < \infty$, and let q be conjugate to p . If X is reflexive, then for every $f \in S_{L^p(\mu, X)}$ the state space*

$$S_f = \left\{ g \in B_{L^q(\mu, X^*)} : \int_{\Omega} g(\omega)(f(\omega)) d\mu(\omega) = 1 \right\}$$

is weakly compact in $L^q(\mu, X^)$.*

Proof. Since X is reflexive, X^* is reflexive. Because $1 < q < \infty$ and μ is finite, the Bochner space $L^q(\mu, X^*)$ is reflexive. Define

$$\Phi_f(g) = \int_{\Omega} g(\omega)(f(\omega)) d\mu(\omega), \quad g \in L^q(\mu, X^*).$$

Holder's inequality gives $\|\Phi_f\| \leq \|f\|_p = 1$, and the usual duality formula gives $\|\Phi_f\| = 1$. Hence

$$S_f = B_{L^q(\mu, X^*)} \cap \Phi_f^{-1}(\{1\}).$$

This is a norm-closed convex bounded subset of the reflexive space $L^q(\mu, X^*)$. Closed convex subsets are weakly closed, so S_f is weakly closed and bounded in a reflexive space. Therefore S_f is weakly compact. $\square$

Proposition 1. *For $p > 1$, weak compactness of S_f does not imply that S_f is a singleton, even when X is two-dimensional.*

Proof. Take $\Omega = [0, 1]$ with Lebesgue measure and $X = (\mathbb{R}^2, \|\cdot\|_1)$, so $X^* = (\mathbb{R}^2, \|\cdot\|_\infty)$. Let $f(\omega) = e_1 = (1, 0)$ for all ω . Then $\|f\|_p = 1$.

If $h \in L^\infty[0, 1]$ satisfies $|h| \leq 1$ almost everywhere, set

$$g_h(\omega) = (1, h(\omega)) \in X^*.$$

Then

$$\|g_h\|_{L^q(\mu, X^*)} = \left(\int_0^1 \max\{1, |h(\omega)|\}^q d\omega \right)^{1/q} = 1$$

and

$$\int_0^1 g_h(\omega)(f(\omega)) d\omega = \int_0^1 1 d\omega = 1.$$

Thus every such g_h belongs to S_f . In particular, g_0 and g_1 are distinct elements of S_f . By the theorem, S_f is nevertheless weakly compact. $\square$

Verification code

The script `code/check_lp_state_space_example.py` checks the finite-partition version of the last calculation for three choices of h . It verifies that the L^q norm and the pairing with the constant e_1 function are both equal to 1.

Scope

This is a partial answer to Problem 3.18. The non-reflexive case remains open in this packet. The main conclusion is that, for $1 < p < \infty$, reflexivity of X makes weak compactness automatic, and the correct L^p theory cannot be a direct copy of the source paper's L^1 singleton theorem.